

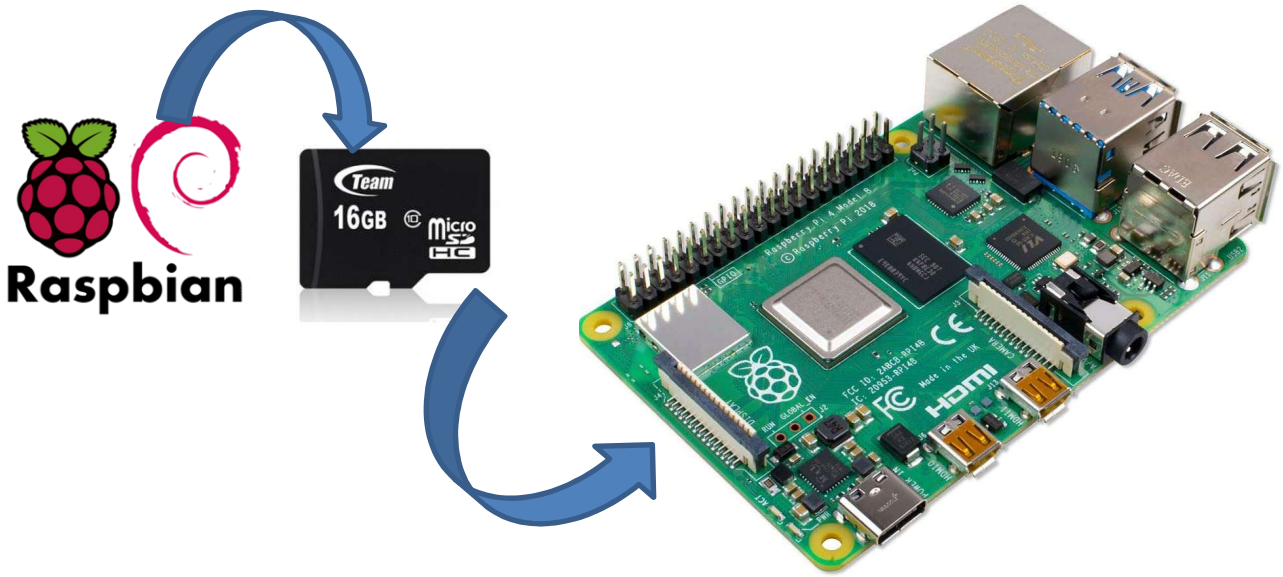
Tanit Training & Consulting

AI & IoT with Raspberry pi and ESP32

Dr. Walid Ben Mabrouk



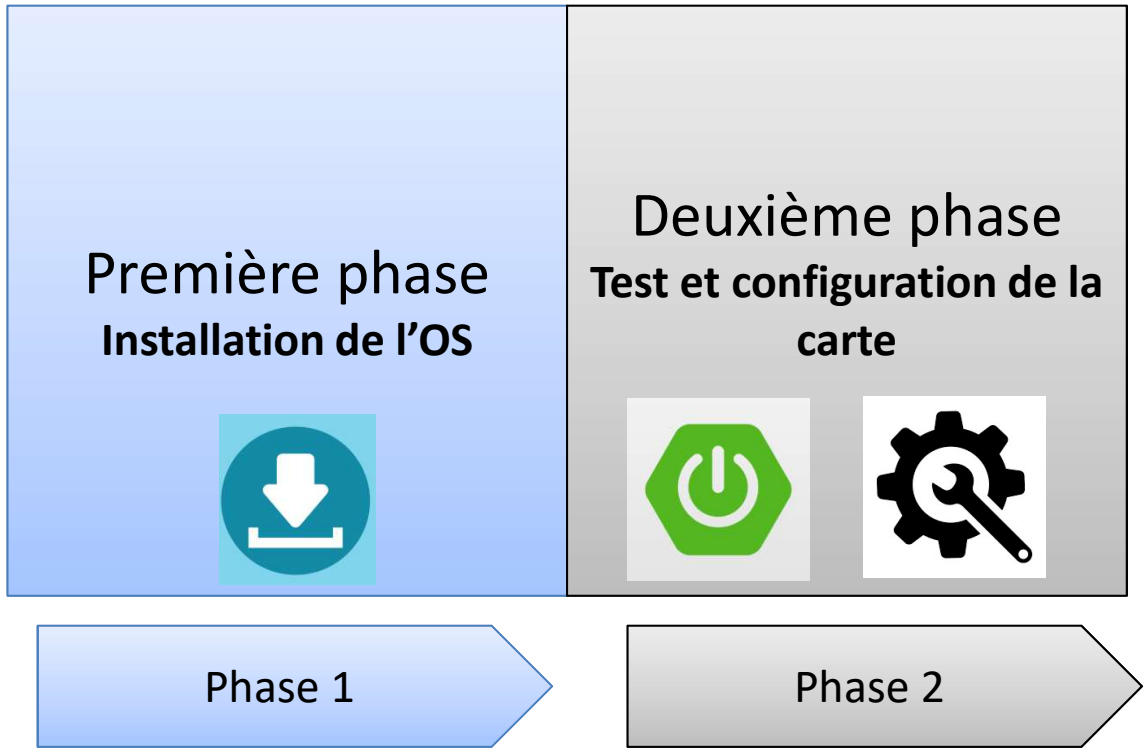

Raspberry Pi
HDMI


Raspbian
Team
16GB
micro SD HC


14:01
ASUS






Première phase
Installation de l'OS
Deuxième phase
Test et configuration de la carte
Phase 1
Phase 2

Télécharger et installer Raspberry pi imager: https://www.raspberrypi.com/software/

Install Raspberry Pi OS using Raspberry Pi Imager

Raspberry Pi Imager is the quick and easy way to install Raspberry Pi OS and other operating systems to a microSD card, ready to use with your Raspberry Pi.

Download and install Raspberry Pi Imager to a computer with an SD card reader. Put the SD card you'll use with your Raspberry Pi into the reader and run Raspberry Pi Imager.

Download for Windows

Download for macOS

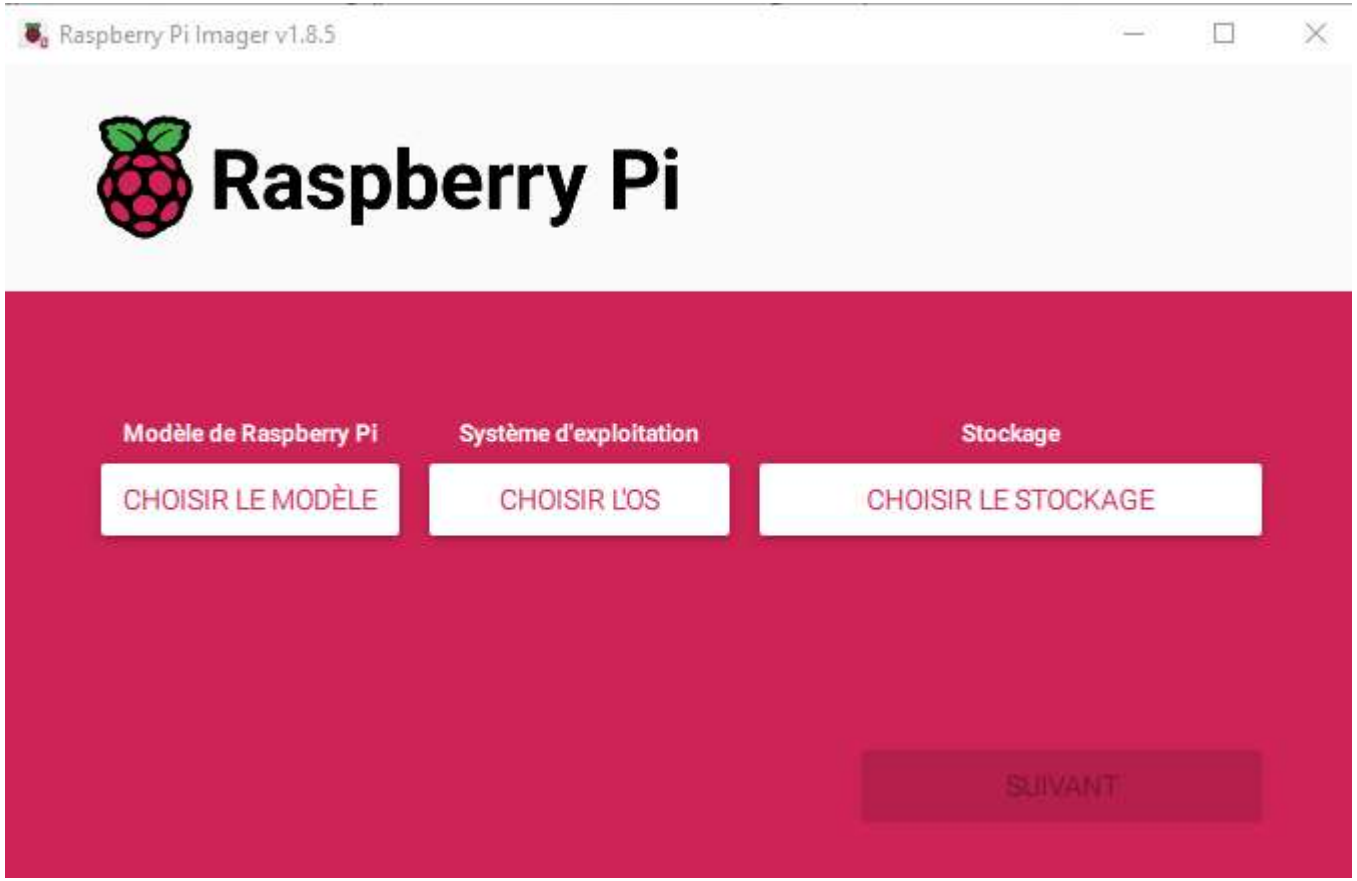
Download for Ubuntu for x86



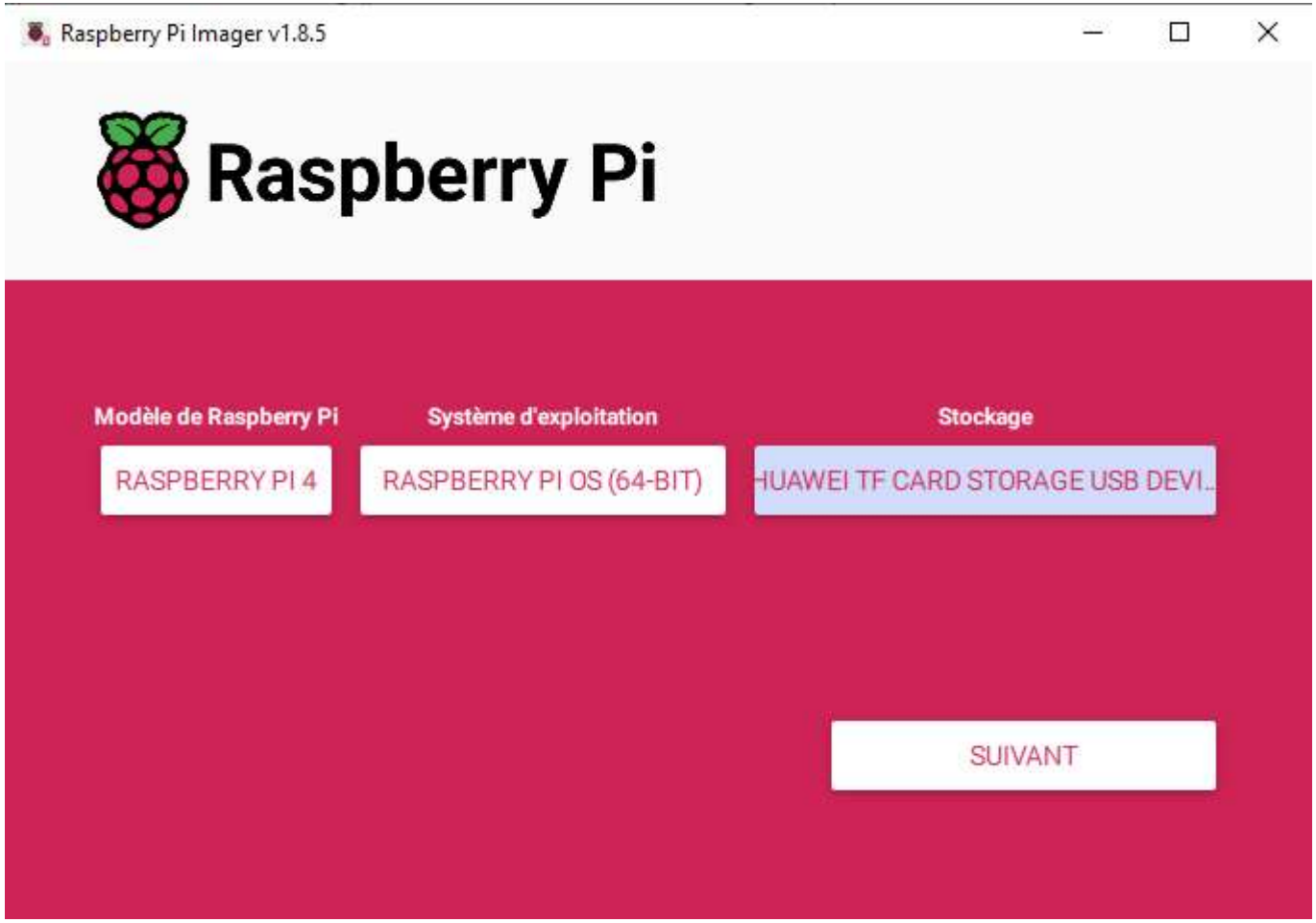
Placer la micro-sd (32G et classe10) dans le PC



Ouvrir l’application Raspberry pi imager

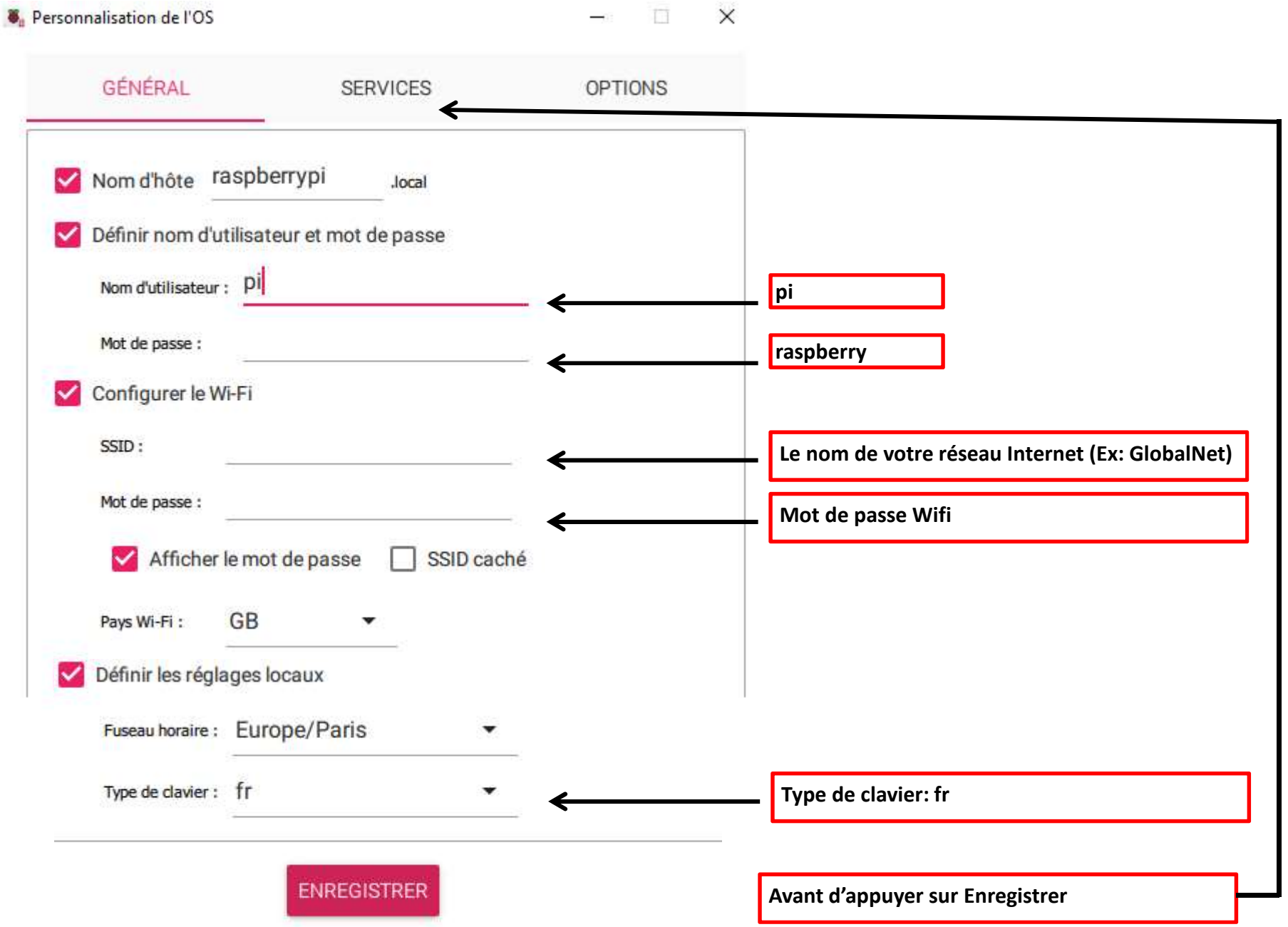


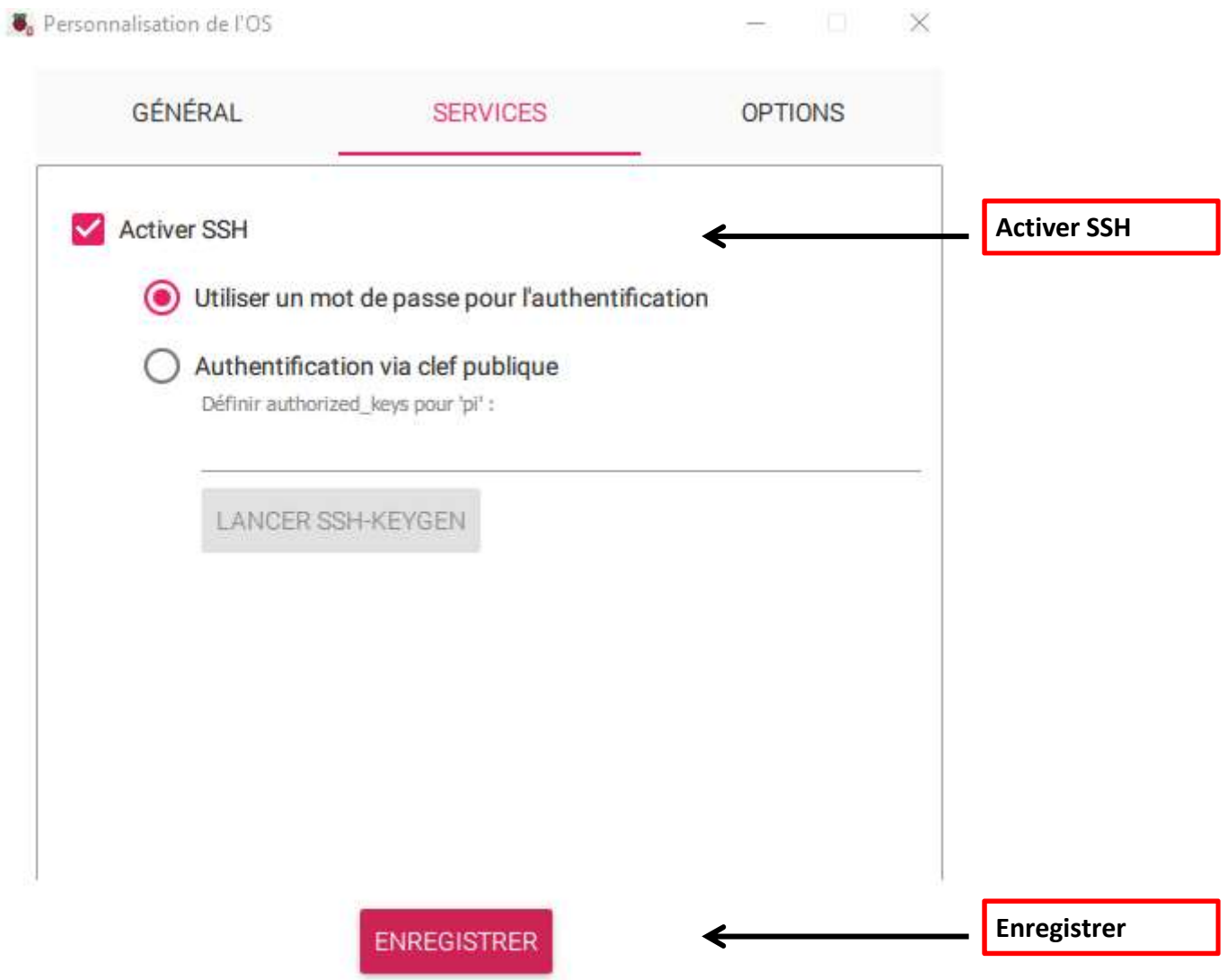
Dans Modèle choisir: **Raspberry pi 4**
Dans système d’exploitation choisir: **Raspberry pi OS (64 bits)**
Dans stockage choisir: **le nom attribué à votre micr-sd**
Appuyer sur suivant

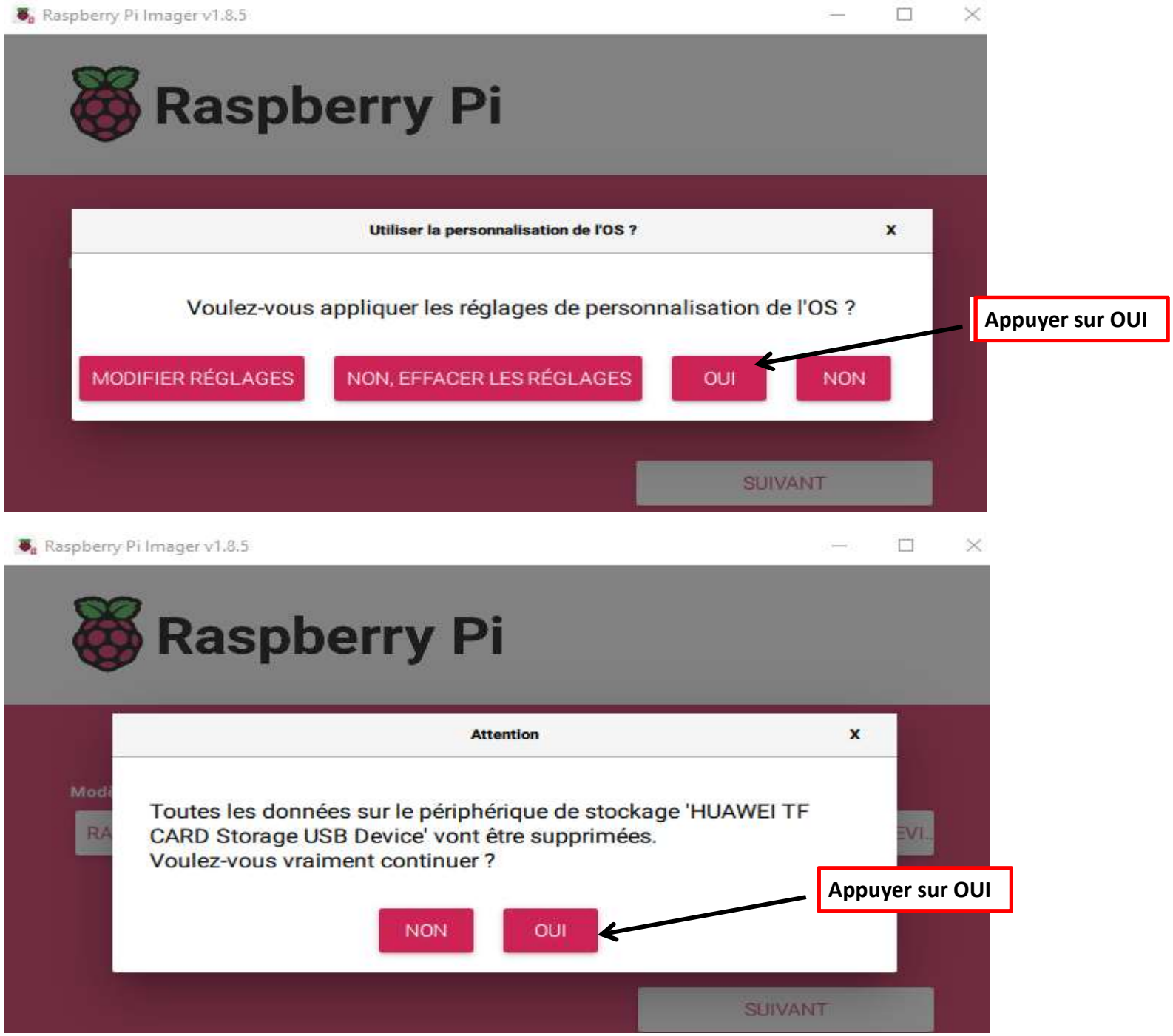


Choisir ‘Modifier Réglages’

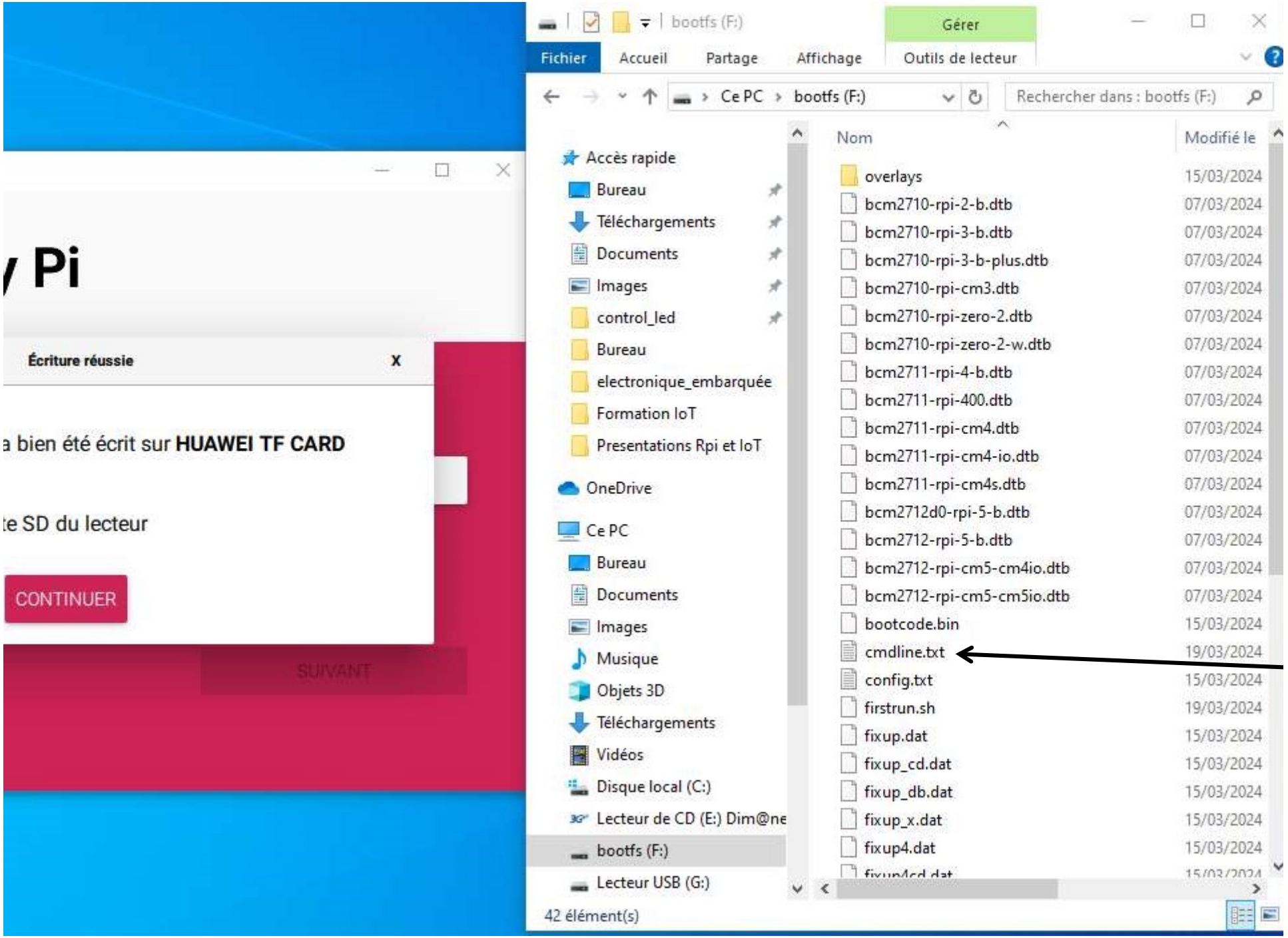


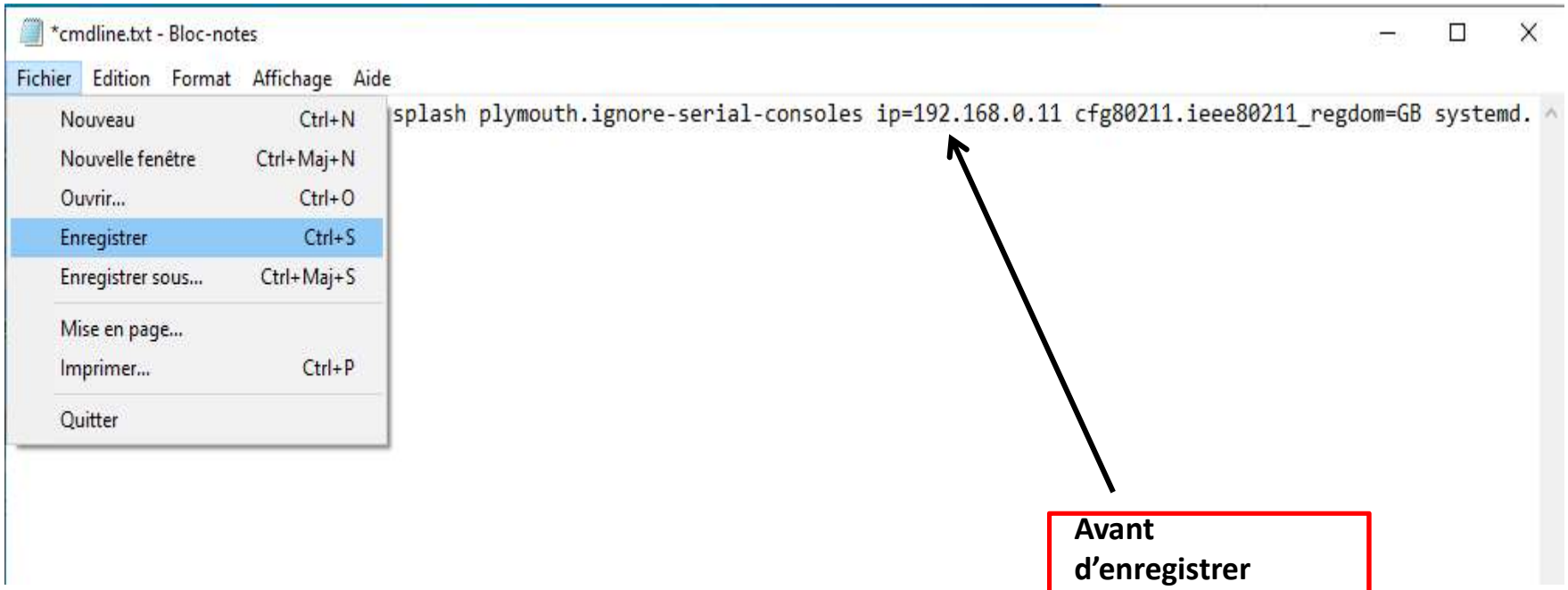

Personnalisation de l'OS
GÉNÉRAL
SERVICES
OPTIONS
Nom d'hôte raspberrypi .local
Définir nom d'utilisateur et mot de passe
Nom d'utilisateur : pi
Mot de passe :
Configurer le Wi-Fi
SSID :
Mot de passe :
Afficher le mot de passe
SSID caché
Pays Wi-Fi : GB
Définir les réglages locaux
Fuseau horaire : Europe/Paris
Type de clavier : fr
ENREGISTRER
pi
raspberry
Le nom de votre réseau Internet (Ex: GlobalNet)
Mot de passe Wifi
Type de clavier: fr
Avant d'appuyer sur Enregistrer


Personnalisation de l'OS
GÉNÉRAL
SERVICES
OPTIONS
Activer SSH
Utiliser un mot de passe pour l'authentification
Authentification via clef publique
Définir authorized_keys pour 'pi' :
LANCER SSH-KEYGEN
ENREGISTRER
Activer SSH
Enregistrer


Raspberry Pi Imager v1.8.5
Raspberry Pi
Utiliser la personnalisation de l'OS ?
Voulez-vous appliquer les réglages de personnalisation de l'OS ?
MODIFIER RÉGLAGES
NON, EFFACER LES RÉGLAGES
OUI
NON
SUIVANT
Appuyer sur OUI
Raspberry Pi Imager v1.8.5
Raspberry Pi
Attention
Toutes les données sur le périphérique de stockage 'HUAWEI TF CARD Storage USB Device' vont être supprimées.
Voulez-vous vraiment continuer ?
NON
OUI
SUIVANT
Appuyer sur OUI


Raspberry Pi Imager v1.8.5
Raspberry Pi
Modèle de Raspberry Pi
Système d'exploitation
Stockage
RASPBERRY PI 4
RASPBERRY PI OS (64-BIT)
HUAWEI TF CARD STORAGE USB DEVI...
Écriture... 2%
ANNULER L'ÉCRITURE


y Pi
Écriture réussie
X
a bien été écrit sur HUAWEI TF CARD
te SD du lecteur
CONTINUER
SUIVANT
bootfs (F:)
Gérer
Fichier
Accueil
Partage
Affichage
Outils de lecteur
Ce PC › bootfs (F:)
Rechercher dans : bootfs (F:)
Accès rapide
Bureau
Téléchargements
Documents
Images
control_led
Bureau
electronique_embarquée
Formation IoT
Presentations Rpi et IoT
OneDrive
Ce PC
Bureau
Documents
Images
Musique
Objets 3D
Téléchargements
Vidéos
Disque local (C:)
Lecteur de CD (E:) Dim@ne
bootfs (F:)
Lecteur USB (G:)
Nom
Modifié le
overlays 15/03/2024
bcm2710-rpi-2-b.dtb 07/03/2024
bcm2710-rpi-3-b.dtb 07/03/2024
bcm2710-rpi-3-b-plus.dtb 07/03/2024
bcm2710-rpi-cm3.dtb 07/03/2024
bcm2710-rpi-zero-2.dtb 07/03/2024
bcm2710-rpi-zero-2-w.dtb 07/03/2024
bcm2711-rpi-4-b.dtb 07/03/2024
bcm2711-rpi-400.dtb 07/03/2024
bcm2711-rpi-cm4.dtb 07/03/2024
bcm2711-rpi-cm4-io.dtb 07/03/2024
bcm2711-rpi-cm4s.dtb 07/03/2024
bcm2712d0-rpi-5-b.dtb 07/03/2024
bcm2712-rpi-5-b.dtb 07/03/2024
bcm2712-rpi-cm5-cm4io.dtb 07/03/2024
bcm2712-rpi-cm5-cm5io.dtb 07/03/2024
bootcode.bin 15/03/2024
cmdline.txt 19/03/2024
config.txt 15/03/2024
firstrun.sh 19/03/2024
fixup.dat 15/03/2024
fixup_cd.dat 15/03/2024
fixup_db.dat 15/03/2024
fixup_x.dat 15/03/2024
fixup4.dat 15/03/2024
42 élément(s)



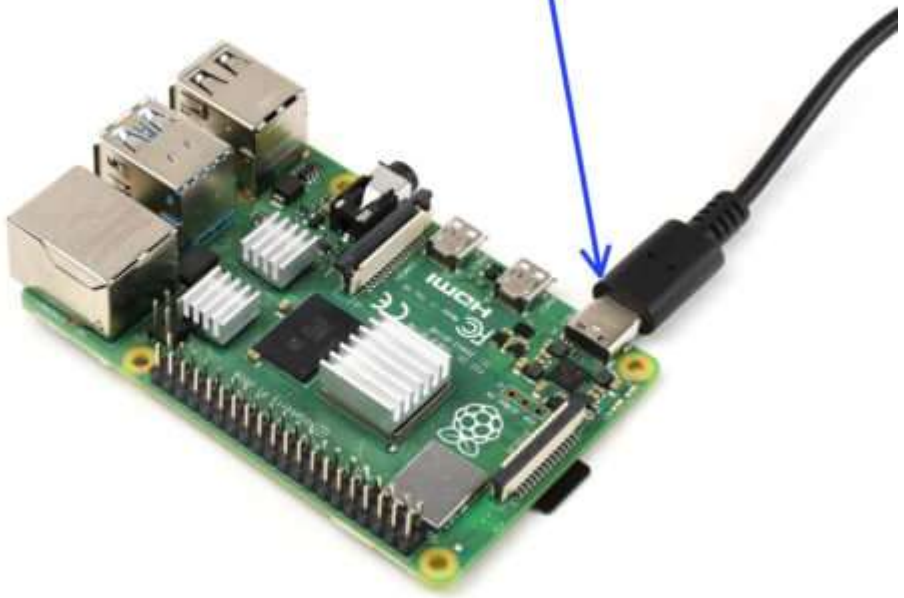
Avant d'enregistrer ajouter: ip=192.168.0.11

Fermer le fichier
Et éjecter la micro-sd

Extraire la µ-SD du PC et la mettre dans la carte Raspberry pi

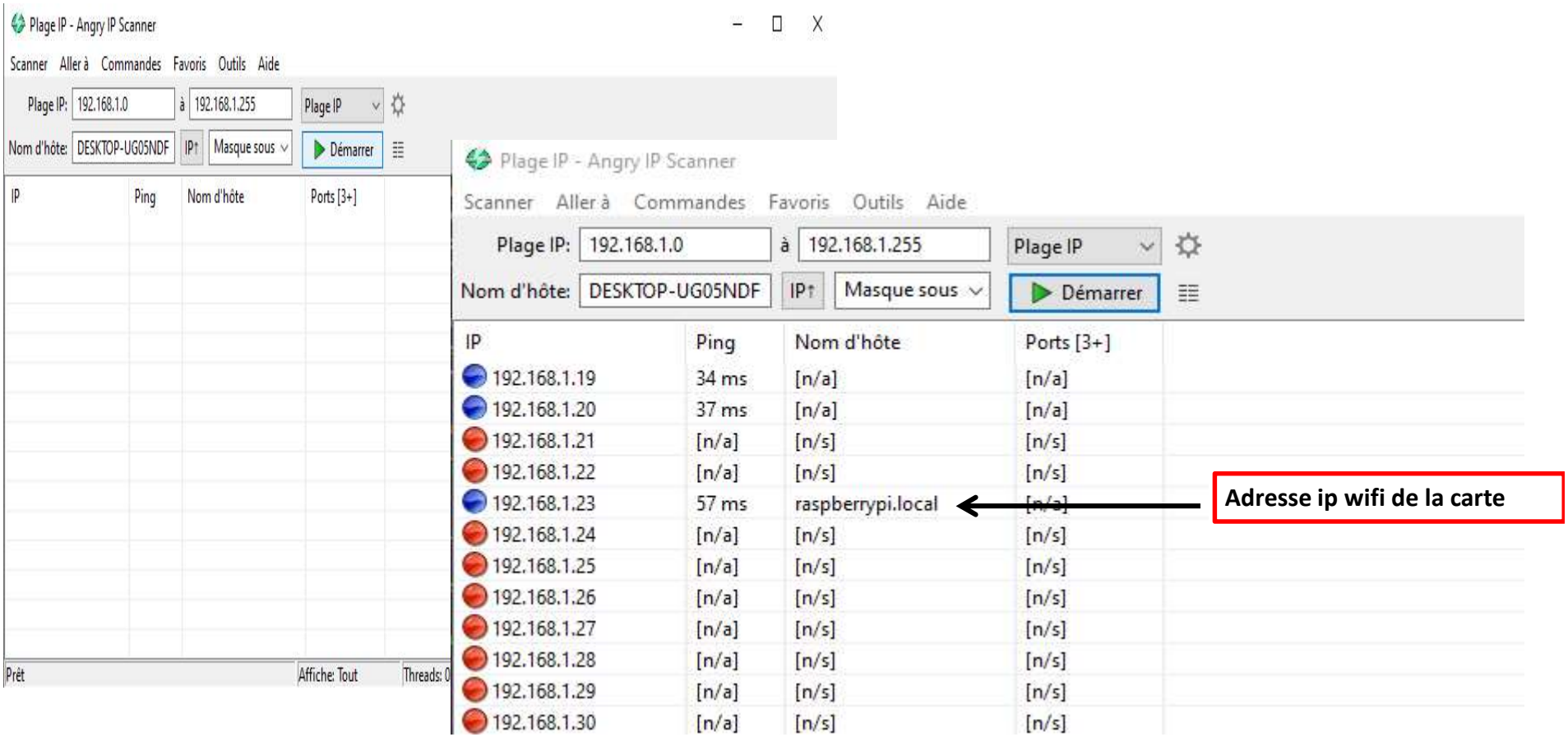


Mettre la carte sous tension



Télecharger et installer l'application putty:
https://www.chiark.greenend.org.uk/~sgtatham/putty/latest.html
Télecharger et installer l'application TigerVNC viewer:
https://sourceforge.net/projects/tigervnc/files/stable/1.13.1/
Télecharger et installer l'application Angry IP scanner:
https://angryip.org/download/#windows

Lancer Angry IP scanner et appuyer sur Démarrer

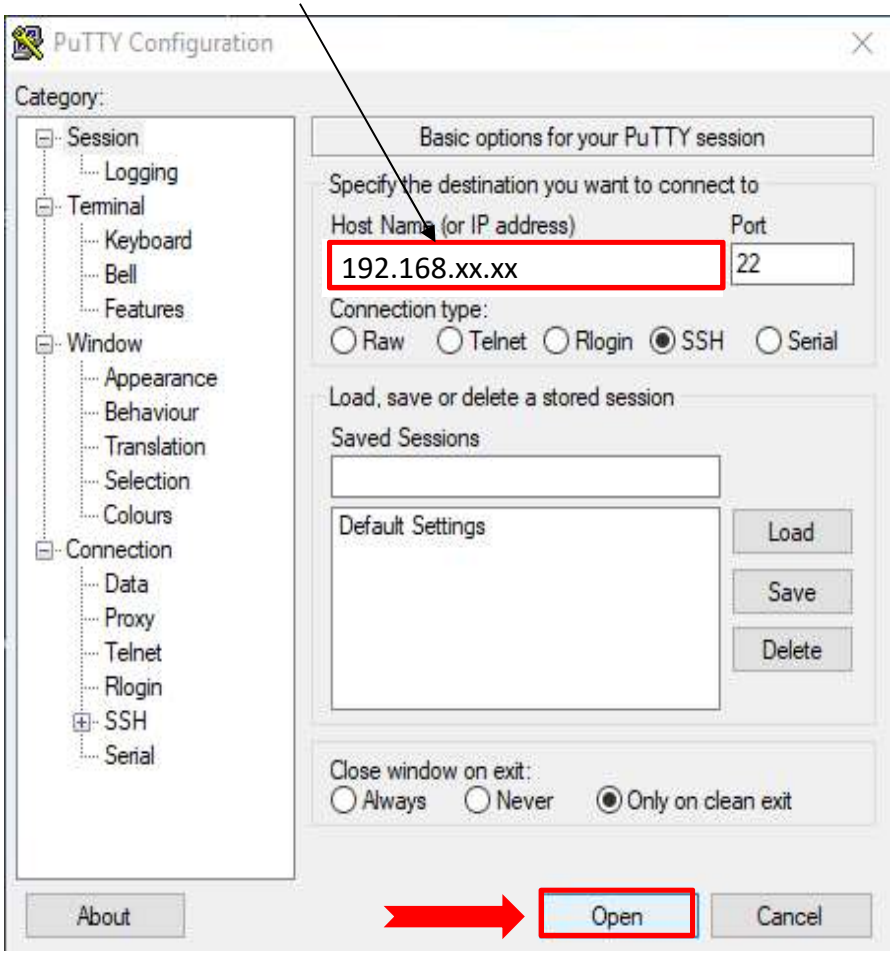


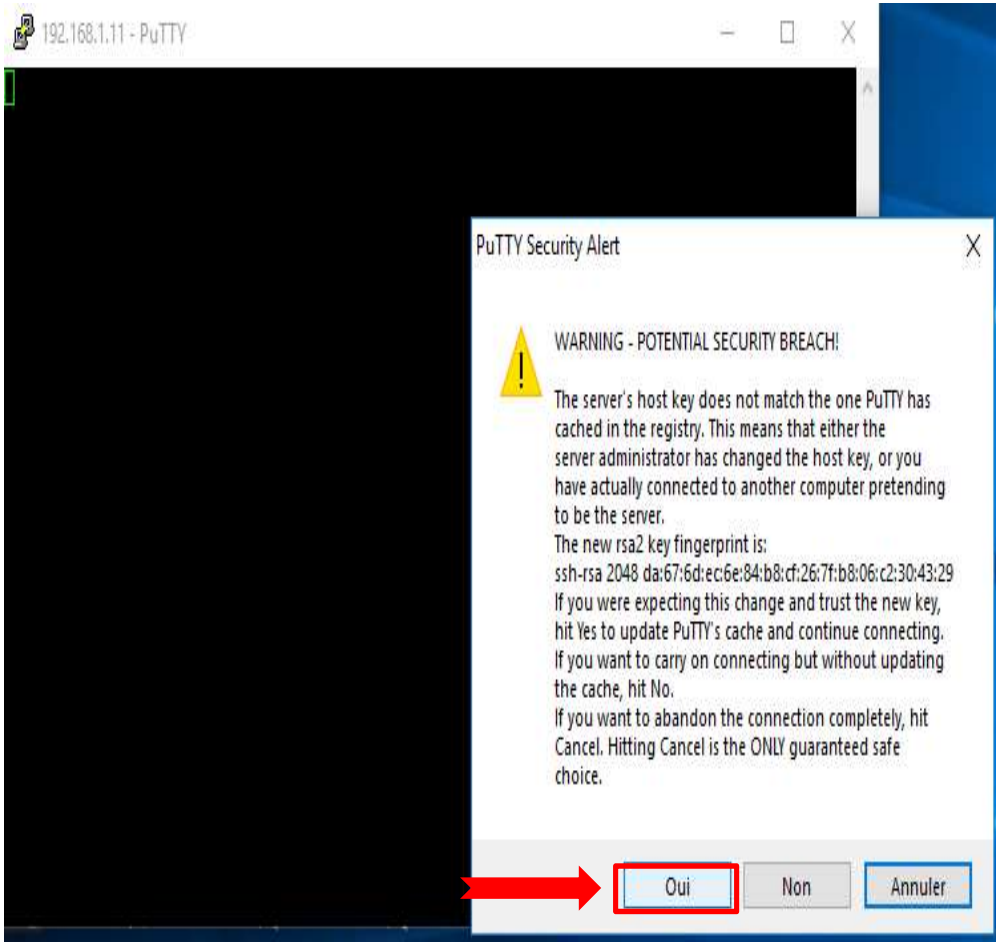
Connexion à la carte

Lancer Putty



Adresse ip wifi de la carte



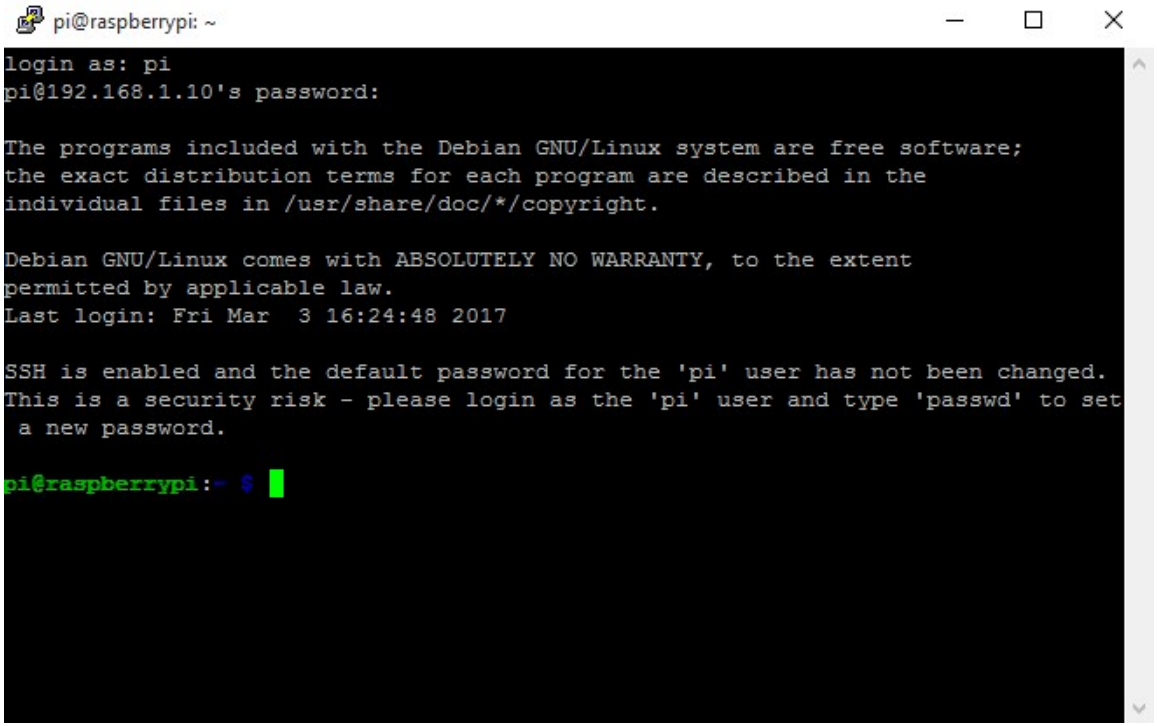


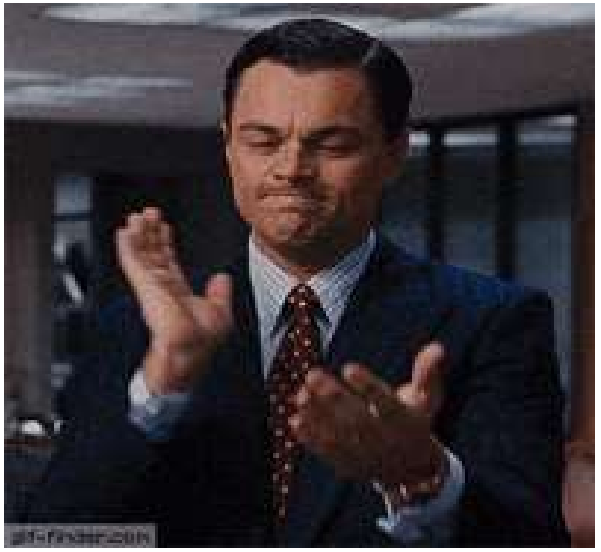
Authentification

Il faut s'authentifier pour se connecter :

login as : pi

password : raspberry



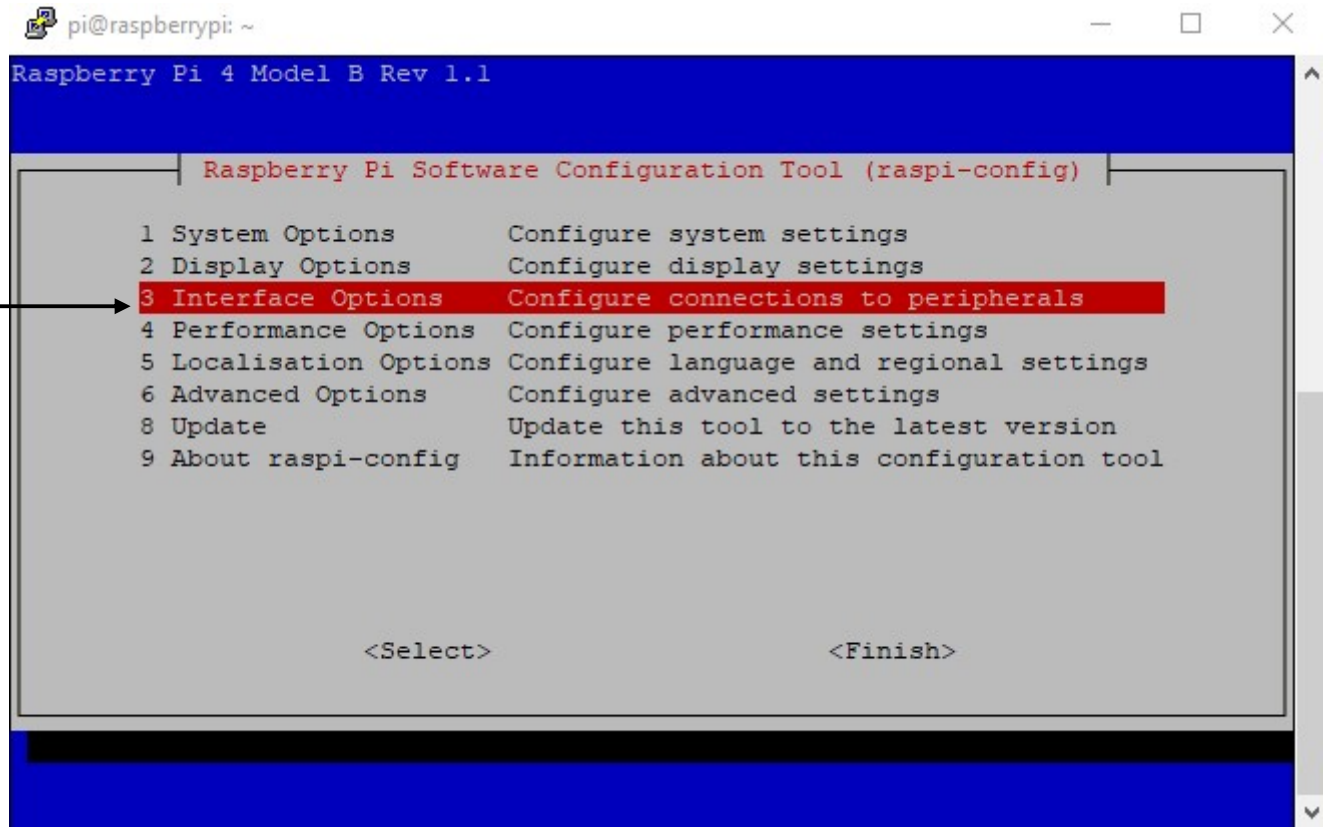


Configuration de la carte

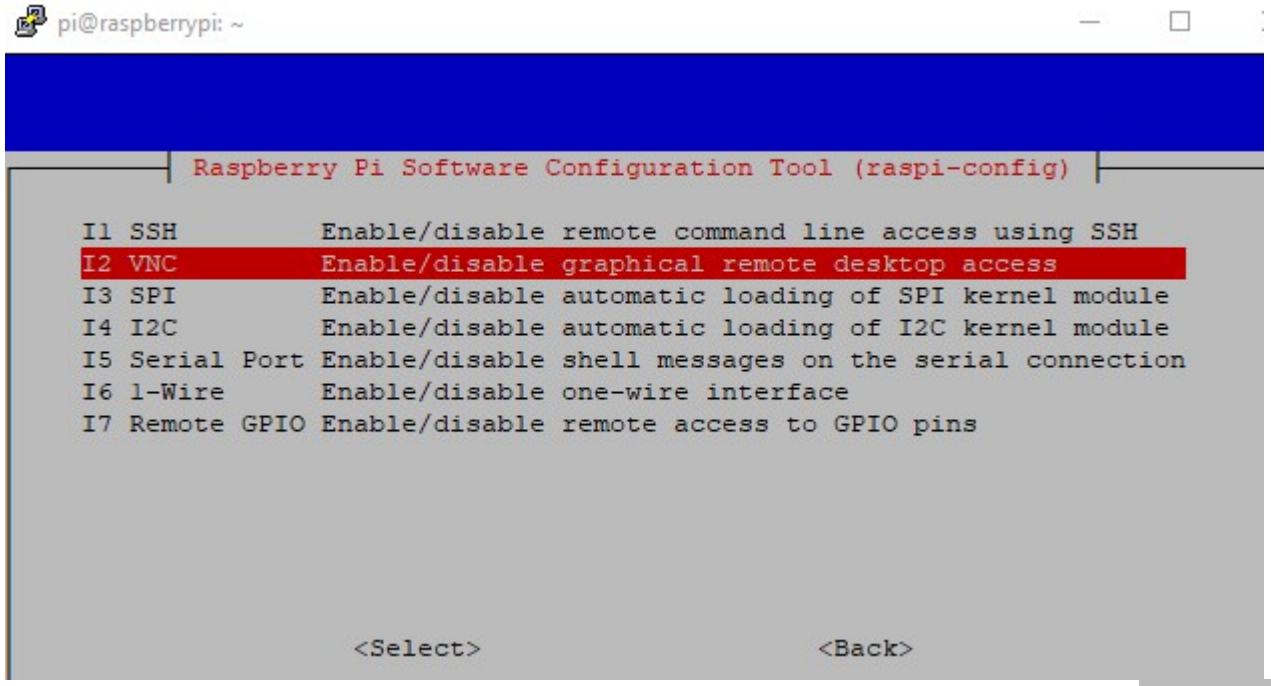
Dans le terminal taper: sudo raspi-config

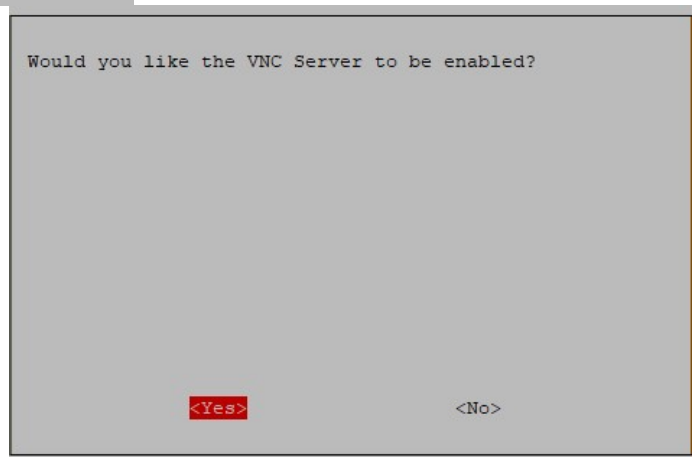
```
pi@raspberrypi:~ $ sudo raspi-config
Created symlink /etc/systemd/system/multi-user.target.wants/vncserver-x11-servic
ed.service → /usr/lib/systemd/system/vncserver-x11-serviced.service.
```

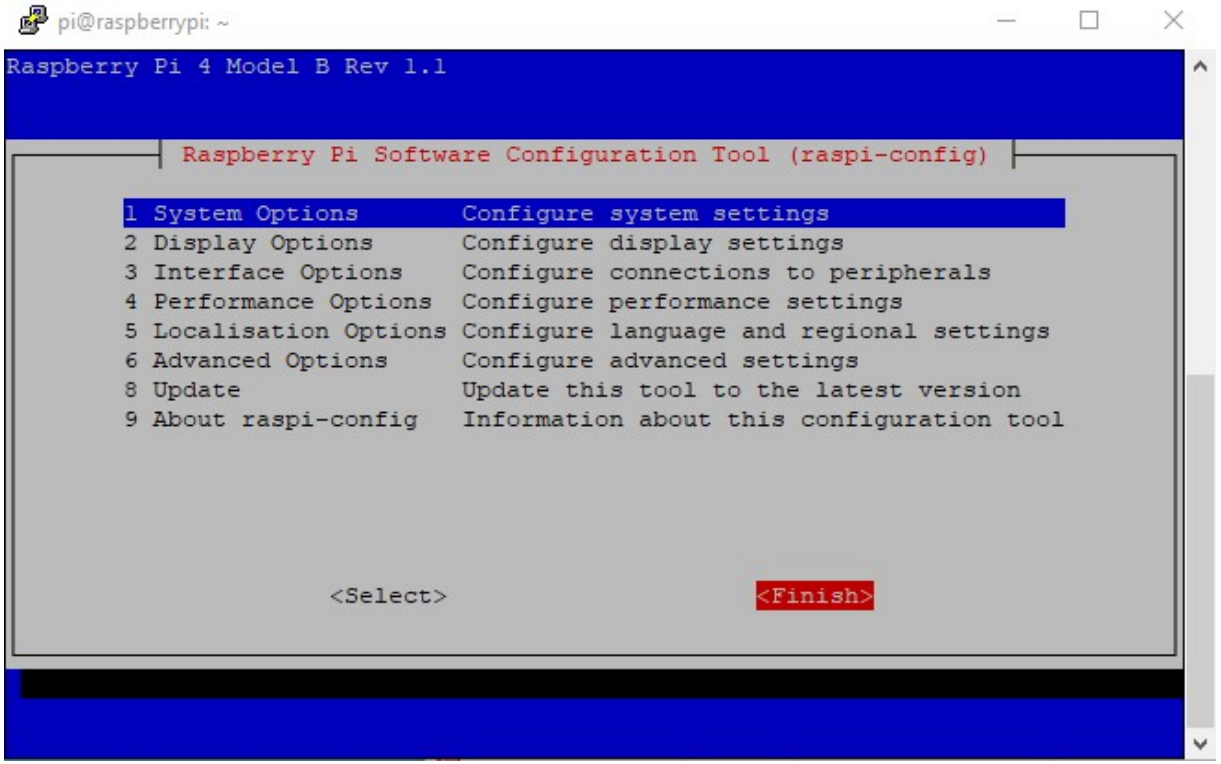
Sélectionner ***Interface options***



Activation de VNC

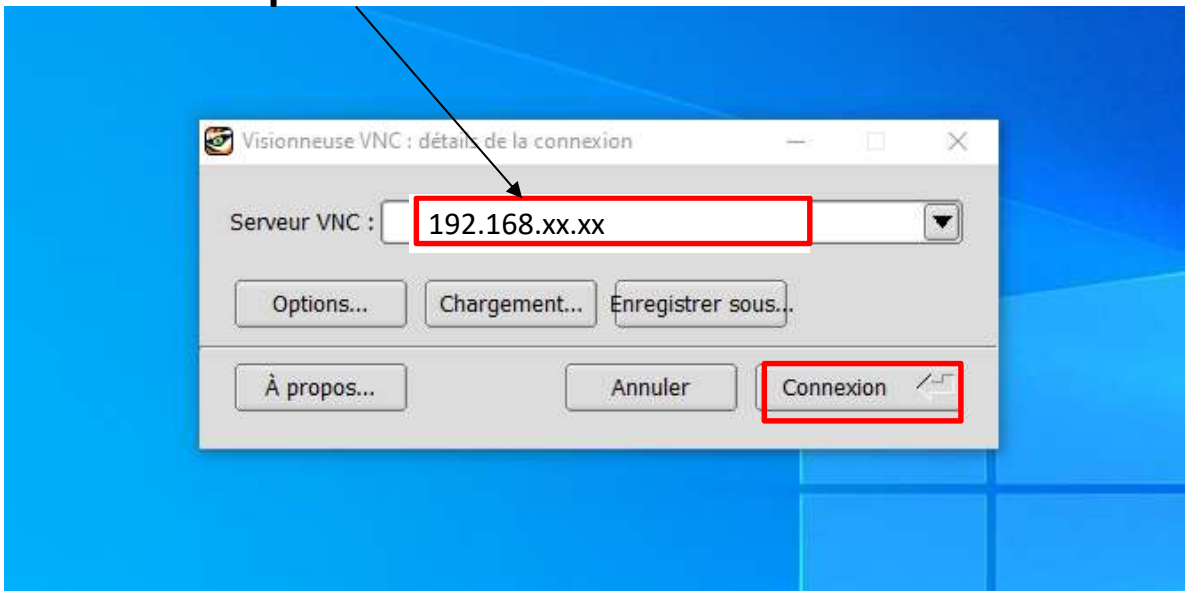




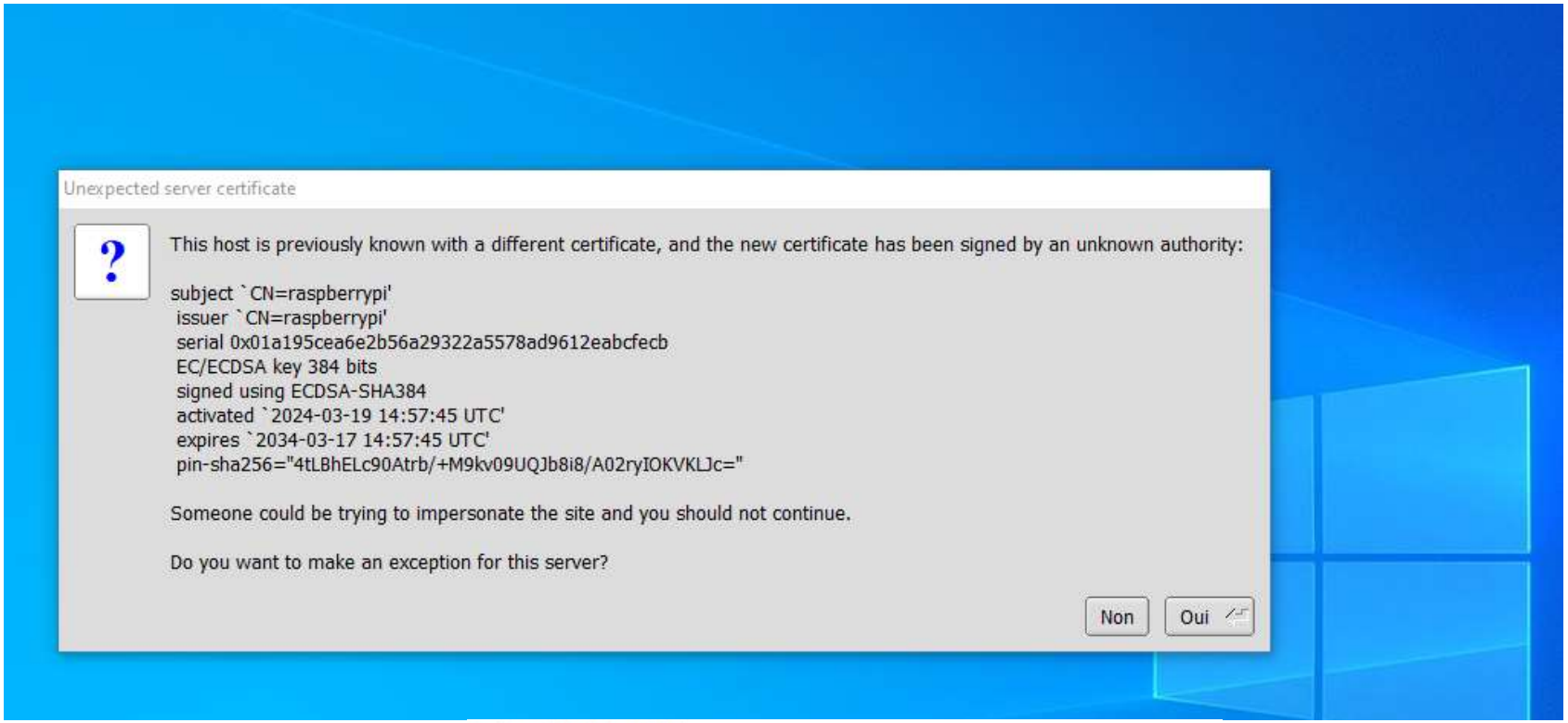


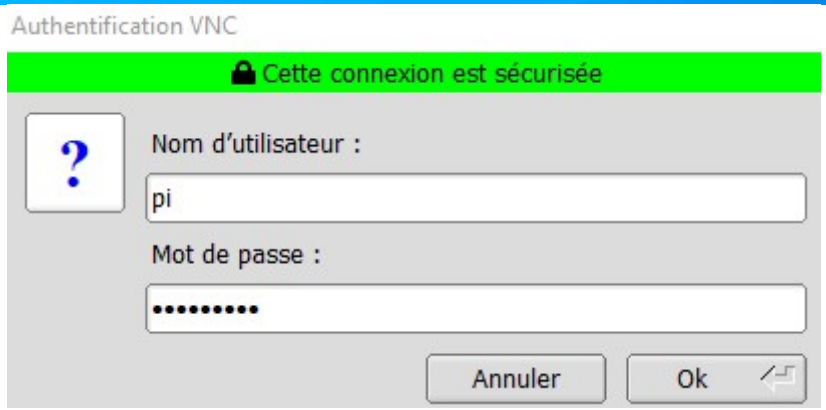
Fermer putty et lancer TigerVNC viewer

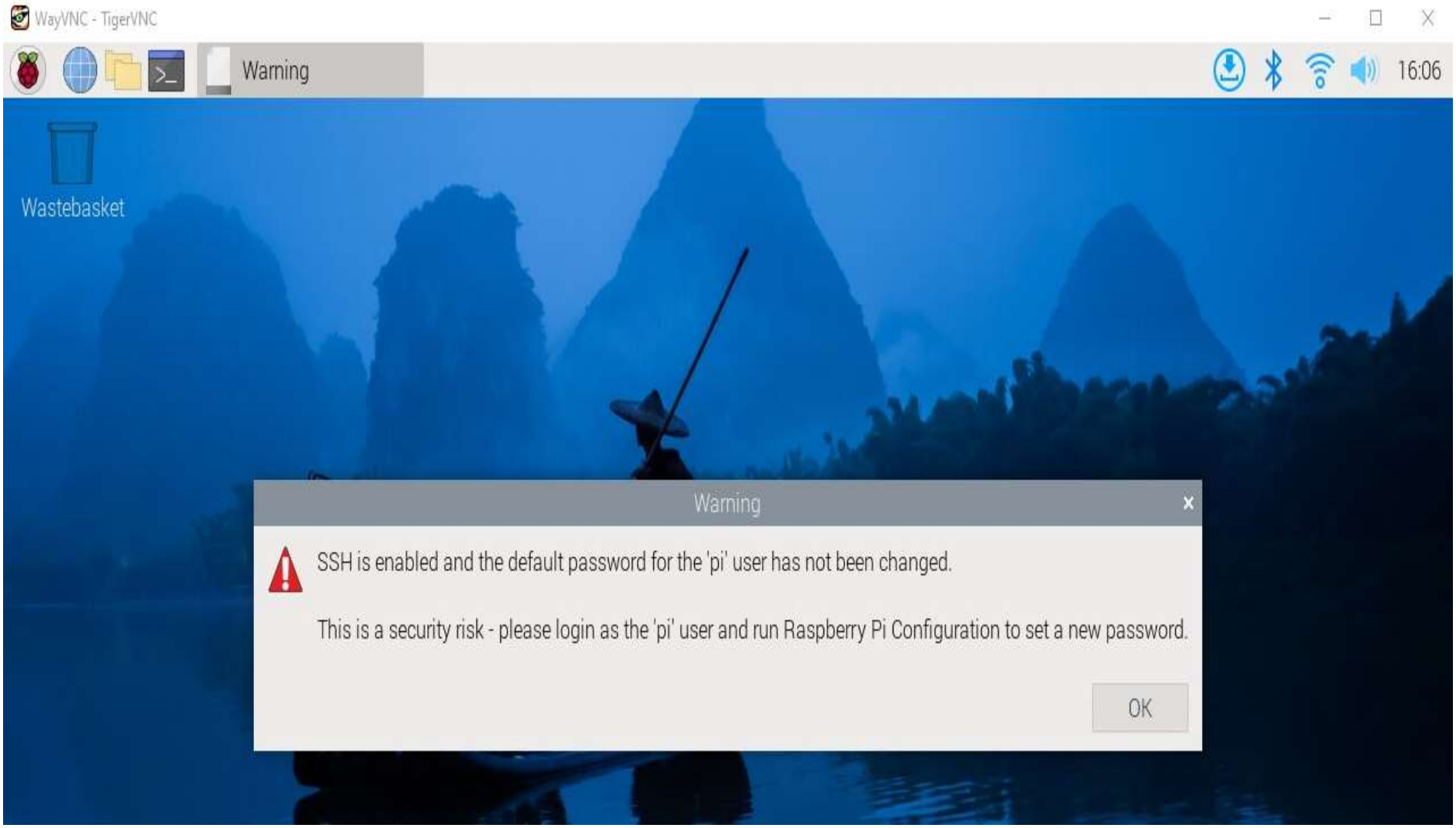
Adresse ip wifi de la carte

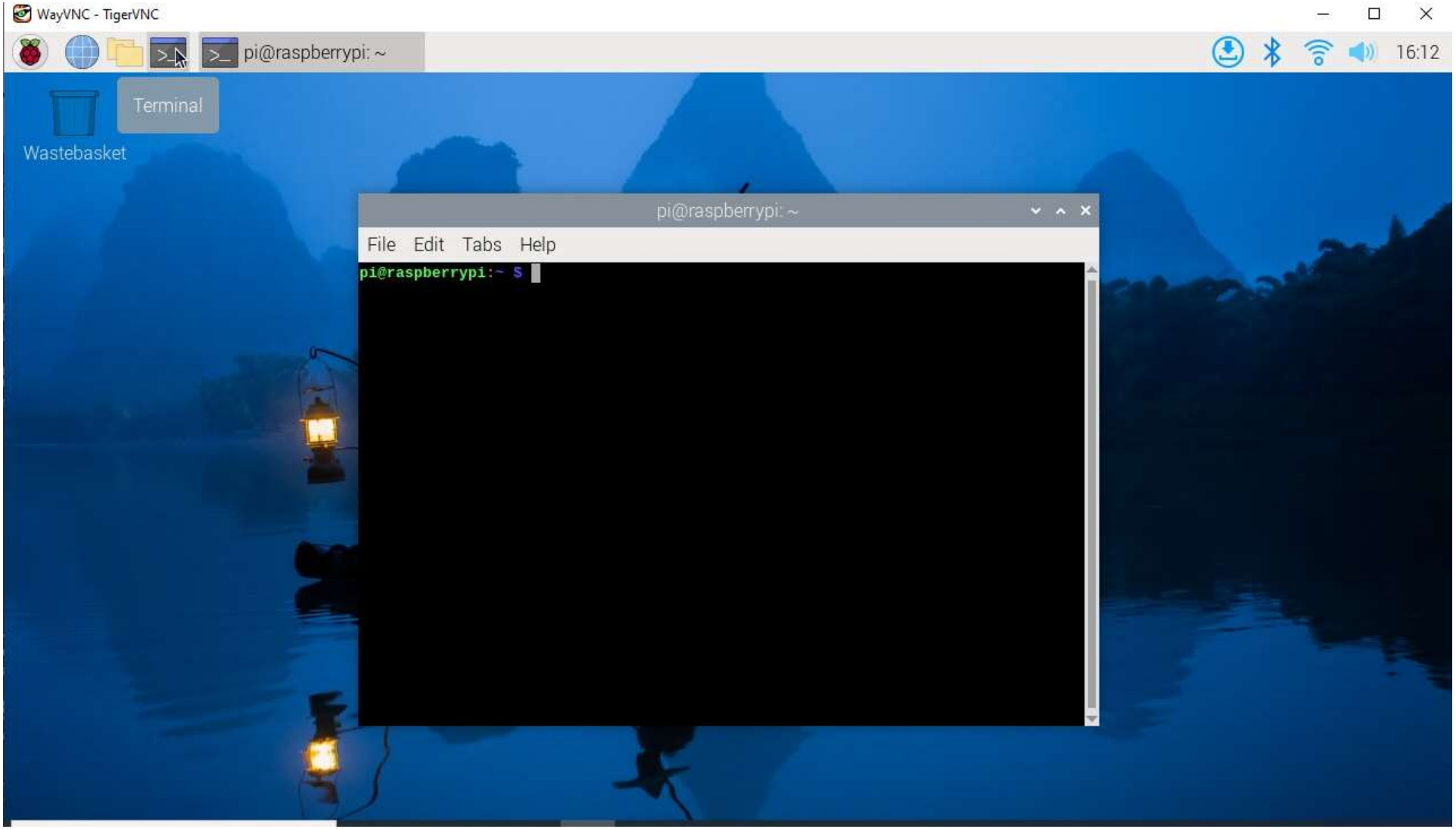



Certificate hostname mismatch
Hostname (192.168.1.23) does not match the server certificate, do you want to continue?
Non
Oui

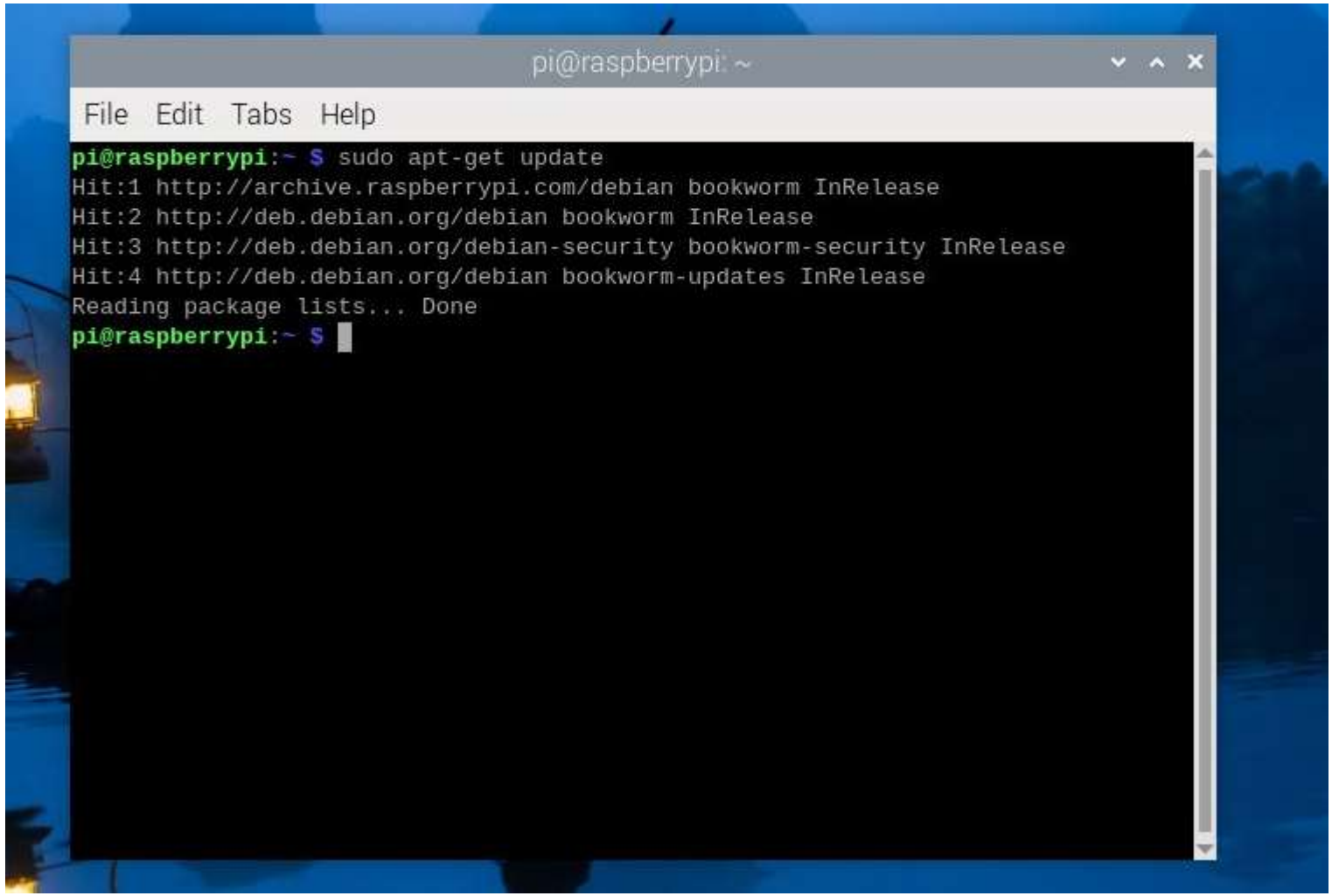

Unexpected server certificate
This host is previously known with a different certificate, and the new certificate has been signed by an unknown authority:
subject `CN=raspberrypi'
issuer `CN=raspberrypi'
serial 0x01a195cea6e2b56a29322a5578ad9612eabcfecb
EC/ECDSA key 384 bits
signed using ECDSA-SHA384
activated `2024-03-19 14:57:45 UTC'
expires `2034-03-17 14:57:45 UTC'
pin-sha256="4tLBhELc90Atrb/+M9kv09UQJb8i8/A02ryIOKVKLJc="
Someone could be trying to impersonate the site and you should not continue.
Do you want to make an exception for this server?
Non
Oui


Authentification VNC
Cette connexion est sécurisée
Nom d'utilisateur :
pi
Mot de passe :
Annuler
Ok


WayVNC - TigerVNC
Warning
16:06
Wastebasket
Warning
SSH is enabled and the default password for the 'pi' user has not been changed.
This is a security risk - please login as the 'pi' user and run Raspberry Pi Configuration to set a new password.
OK


WayVNC - TigerVNC
pi@raspberrypi: ~
16:12
Terminal
Wastebasket
pi@raspberrypi: ~
File Edit Tabs Help
pi@raspberrypi:~ $

Taper: **sudo apt-get update**



Taper et executer ligne par ligne dans le terminal:

sudo apt-get install git python3-pip
git clone https://github.com/Qengineering/Tensorflow-io.git

sudo mv /usr/lib/python3.11/EXTERNALLY-MANAGED /usr/lib/python3.11/EXTERNALLY-MANAGED.old

cd Tensorflow-io
sudo -H pip3 install tensorflow_io_gcs_filesystem-0.23.1-cp311-cp311-linux_aarch64.whl
cd
sudo -H pip3 install --upgrade tensorflow==2.14.0
pip3 install matplotlib
pip3 install scikit-learn
pip3 install pandas
pip3 install opencv-contrib-python

bash <(curl -sL https://raw.githubusercontent.com/node-red/linux-installers/master/deb/update-nodejs-and-nodered)